Physics A (H156, H556)

Exploring - Magnetic fields ppq
KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **30**.
- The total number of marks may take into account some 'either/or' question choices.

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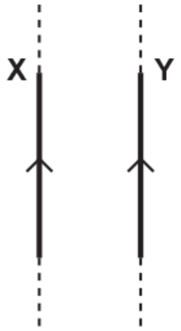
1 The diagram below shows two long current-carrying conductors X and Y.



The conductors are parallel to each other.
Y experiences a force because it is in the magnetic field of X.

Which row gives the correct direction of the magnetic field at Y due to X, and the direction of the force experienced by Y due to this field?

	Direction of magnetic field	Direction of force
A	Down into the plane of paper	To the right
B	Up from the plane of paper	To the right
C	Down into the plane of paper	To the left
D	Up from the plane of paper	To the left

Your answer **[1]**

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2(a) The diagram below shows the top-view of a long current-carrying wire.

The direction of the current in the wire is into the plane of the paper.

Draw at least **three** field lines to indicate the magnetic field pattern around this wire.

[2]

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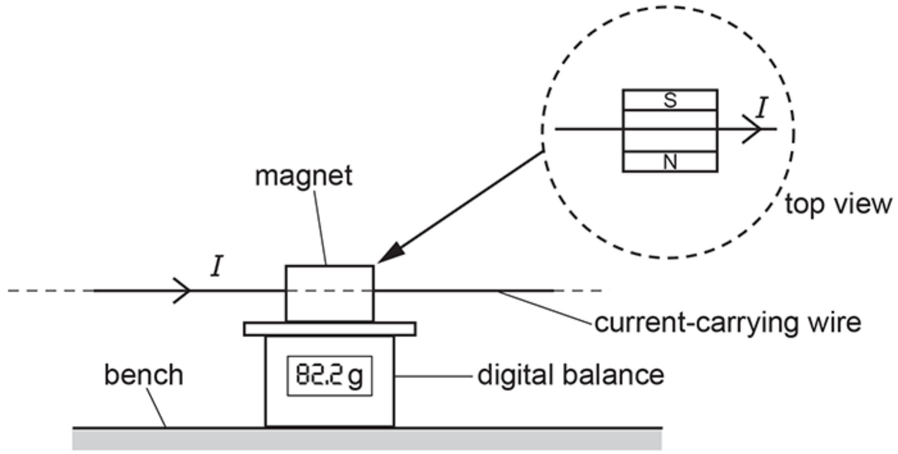
(b) The arrangement shown in the diagram below is used to determine the magnetic flux density between the poles of a permanent magnet



The magnet is placed on the digital balance. The current-carrying wire is horizontal and at right angles to the magnetic field between the poles of the magnet. The wire is fixed.

The following results are collected.

- length of the wire in the uniform field of the magnet = 6.0 ± 0.2 cm
- balance reading with no current in wire = 80.0 g
- balance reading with current in wire = 82.2 g
- current in wire = 5.0 ± 0.1 A

Calculate the magnetic flux density *B*, including the absolute uncertainty.
Ignore the absolute uncertainty in the balance readings.
Write your value for *B* to **2** significant figures and the absolute uncertainty to **1** significant figure.

B = ... ± .... T **[4]**

3 Faraday's law of electromagnetic induction is written below with two terms missing.

The .... induced in a circuit is directly proportional to the rate of change of

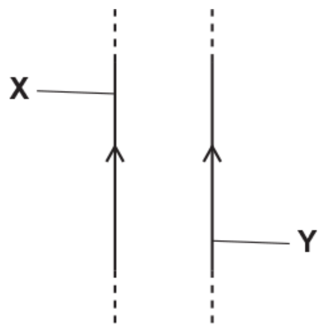
magnetic flux ....

What are the **two** missing terms?

A current, density
B current, linkage
C electromotive force, density
D electromotive force, linkage

Your answer [ ] **[1]**

4 The diagram below shows two long vertical current-carrying wires **X** and **Y**.



The direction of the current in each wire is the same.

Explain why wire **Y** experiences a force and deduce the direction of this force.

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[3]

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5 A student is doing an experiment on the magnetic force experienced by a current-carrying wire in a uniform magnetic field. The magnetic flux density *B* can be varied.

For a particular flux density, the current in the wire is 2.0A. The length of the wire in the field is 0.12 m. The angle between the current and the magnetic field is 30°. The force experienced by the wire is 7.7×10^{-2} N.

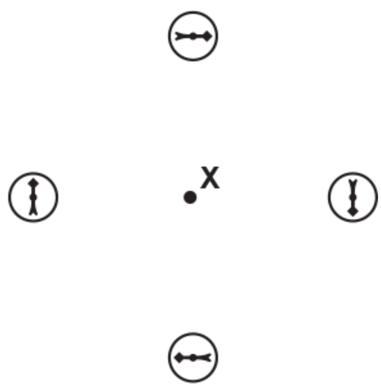
The student calculates *B* and records the results in a table.

Which row shows the correct table heading for *B* and the correct value for *B*?

	Table heading for *B*	Value for *B*
A	*B* / T	0.37
B	*B* / T	0.64
C	*B* / Wb	0.37
D	*B* / Wb	0.64

Your answer [1]

6 The diagram shows four magnetic compasses placed at the same distance from point **X**.



Which of the following is most likely to be at point **X**?

A permanent magnet

B current-carrying solenoid

C current-carrying flat coil

D straight current-carrying wire

Your answer [1]

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7(a)

A student conducts an experiment to confirm that the uniform magnetic flux density B between the poles of a magnet is 30 mT.

A current-carrying wire of length 5.0 cm is placed perpendicular to the magnetic field.

The current I in the wire is changed and the force F experienced by the wire is measured. Fig. 22.1 shows the graph plotted by the student.

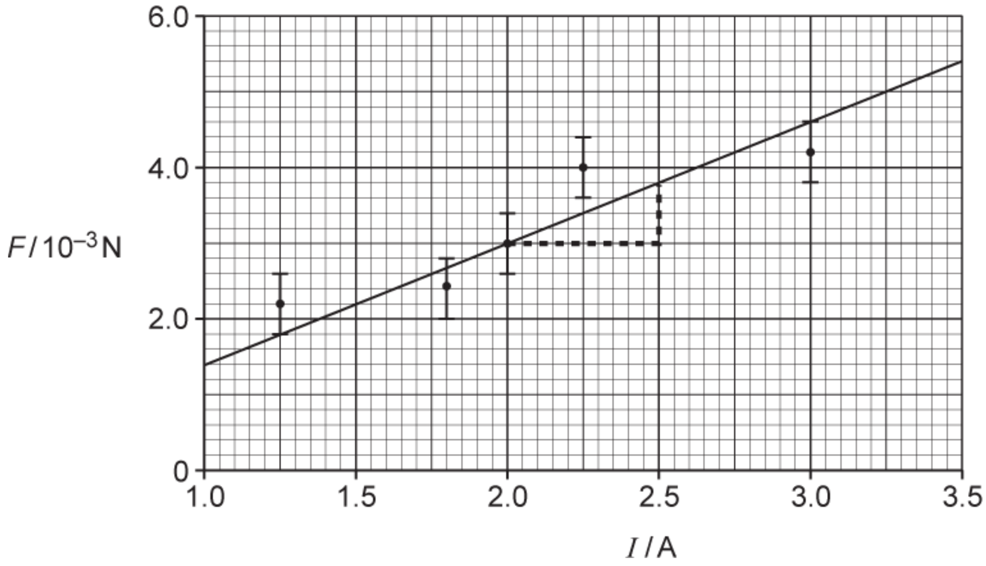

Fig. 22.1

The student's analysis is shown on the graph of Fig. 22.1 and in the space below.

$F = BIL$

$$\text{gradient} = BL = \frac{(3.8 - 3.0) \times 10^{-3}}{2.5 - 2.0} = 0.0016$$

$$B = \frac{0.0016}{0.05} = 0.032\ \text{T} = 32\ \text{mT}$$

This is just 2 mT out from the 30 mT value given by the manufacturer, so the experiment is very accurate.

Evaluate the information from Fig. 22.1 and the analysis of the data from the experiment. No further calculations are necessary.

[6]

(b) Fig. 22.2 shows a transformer circuit.

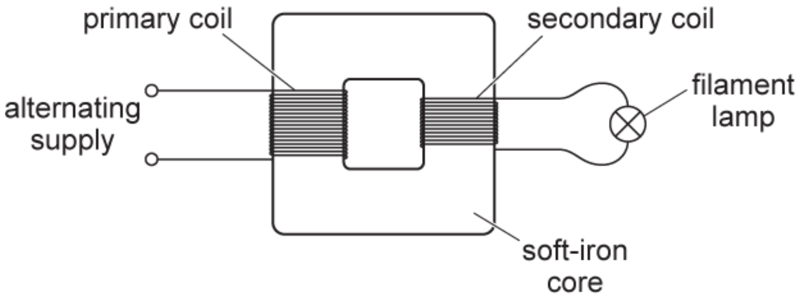

Fig. 22.2

The primary coil is connected to an alternating voltage supply. A filament lamp is connected to the output of the secondary coil.

i. Use Faraday's law of electromagnetic induction to explain why the filament lamp is lit.

[3]

ii. The primary coil has 400 turns and the secondary coil has 20 turns. The potential difference across the lamp is 12 V and it dissipates 24 W. The transformer is 100% efficient.

1. Calculate the current in the primary coil.

current = ___________________ A [2]

2. The alternating voltage supply is replaced by a battery and an open switch in series. The switch is closed. The lamp is lit for a short period of time and then remains off. Explain this observation.

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____

____

____ [2]

8 A nucleus of hydrogen-3 ($^{3}_{1}H$) is unstable and it emits a beta-minus particle (electron).

The emitted beta-minus particle enters a region of uniform magnetic field.
Fig. 22.1 shows the path of the particle **before** it enters the magnetic field.

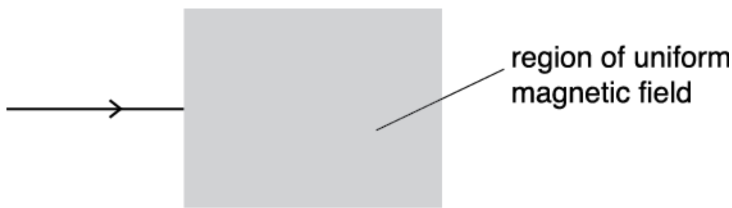

Fig. 22.1

The direction of the magnetic field is into the plane of the paper.
Describe and explain the path of the particle in the magnetic field.

____

____

____

____ [2]

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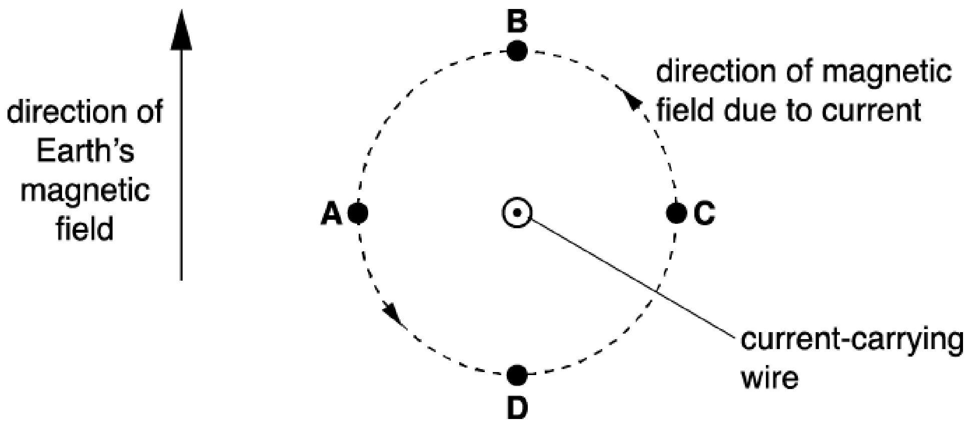
9 The diagram below shows a current-carrying wire coming out from the plane of the paper. The current in the wire produces a magnetic field in an **anticlockwise** direction around the wire.



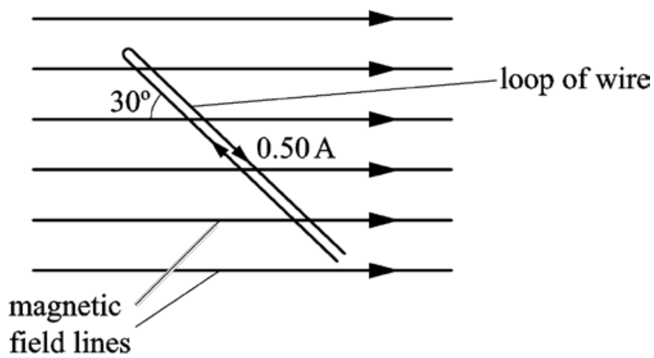
The direction of the Earth's magnetic field is also shown.
The Earth's magnetic field interacts with the magnetic field of the current-carrying wire.

At which point **A**, **B**, **C** or **D** is the resultant magnetic field strength a **minimum**?

Your answer

[1]

10 A rigid loop of insulated wire is placed in a uniform magnetic field of flux density 80 mT. The current in this loop is 0.50 A and the angle between the wire and the direction of the magnetic field is 30°.



What is the magnitude of the force experienced by a 1.0 cm section of the loop?

A. 0 N
B. 2.0×10^{-4} N
C. 3.5×10^{-4} N
D. 4.0×10^{-4} N

Your answer

[1]

END OF QUESTION PAPER

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Question			Answer/Indicative content	Marks	Guidance
1			C	1	<u>Examiner's Comments</u> Most candidates appear to know that the force between the two wires would be attractive, and circled C or D as their options. A few used the diagram and the right hand rule to draw arrows to help with the direction. Both of these are good practice in this style of question. A little less than half of the candidates selected the correct response with the vast majority of incorrect responses being D.
			Total	1	
2	a		Direction of field shown as clockwise <u>Three</u> field lines shown as concentric circles and distance between adjacent field lines increasing as distance from wire increases	B1 B1	Expect at least one field line with an arrow **Allow** more than three lines, but distance between adjacent field lines increasing distance from wire must increase for all <u>Examiner's Comments</u> This question requires the candidates to identify the direction of the field and also to appreciate that the magnitude of the field reduces as the distance from the wire increases. Only around half were able to apply the right hand rule correctly to determine the direction, and only around 10% scored both marks. The increasing separation of the field lines with distance was poorly done for the most part. Many candidates kept the same separation, however those that may have attempted to increase this did not do with any clarity, so that parts of the circle would decrease. In general, the quality of the circles meant that it was difficult to be sure what the candidate's intention was. Some candidates were confused by the leader line, thinking it was the wire and attempted to draw a pattern around this. The question is clear that the diagram represents a top-view.

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Question			Answer/Indicative content	Marks	Guidance
	b		(force =) $2.2 \times 10^{-3} \times 9.81$ $2.2 \times 10^{-3} \times 9.81 = B \times 5.0 \times 0.060$ (= 0.072 T) (absolute uncertainty =) $\frac{0.2}{6.0} + \frac{0.1}{5.0}$ (×0.072 = 0.0038 T) $B = 0.072 \pm 0.004$	C1 C1 C1 A1	**Allow** calculation of percentage uncertainty = 5.3% **Allow** calculation of max B (=0.0759 T) and min B (=0.0683 T) **Note** *B* must be given to 2 SF and the uncertainty given to 1 SF. **Special case:** allow follow through from incorrect B calculation. <u>Examiner's Comments</u> This question is based around a common experiment used to determine the magnetic flux density of a pair of magnets and the experimental design should have been familiar to many candidates, along with the use of $F = BIL\sin\theta$ from the data booklet. The first mark is for identifying the magnitude of the force as being the change in the apparent weight on the balance. Several candidates simply used the reading with the wire, or did not change the mass unit to kg. However, those who managed to get the correct reading for the force generally went on to calculate the magnetic flux density correctly. The uncertainties for two readings were given, and most candidates correctly calculated a percentage uncertainty of 5.3%. The final answer required the correct number of significant figures. Some candidates either did not see this, or ignored it, leaving their final answer in different significant figures. It was noted that several candidates underlined this instruction and in general they tended to follow it. It is good practice to do this.
			Total	6	

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Question			Answer/Indicative content	Marks	Guidance
3			D	1	Examiner's Comments The correct response is **D**. This question was correctly answered by the majority of candidates, although almost all the incorrect responses were **C**, presumably as candidates are aware that it is the e.m.f. that is induced but less familiar with Faraday's law in general.
			Total	1	
4			Magnetic field (around current-carrying wire) (Fleming's) left-hand rule mentioned (Magnetic) field into page, (current is up the page) **and** force is to the left / towards **X**	B1 B1 B1	**Not** magnetic force **Allow** 'field into page and wires attract' **Note** the field direction and force direction can be shown on the figure
			Total	3	
5			B	1	
			Total	1	
6			D	1	
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
7	a		**Level 3 (5-6 marks)** Clear evaluation of Fig. 22.1 **and** clear analysis *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3-4 marks)** Some evaluation of Fig. 22.1 **and** some analysis *There is a line of reasoning presented with some structure.* *The information presented is in the most part relevant and supported by some evidence.* **Level 1 (1-2 marks)** Limited evaluation of Fig. 22.1 **or** limited analysis *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.* **0 marks** *No response or no response worthy of credit.*	B1×6	Use level of response annotations in RM Assessor, e.g. L2 for 4 marks, L2ˆ for 3 marks, etc. **Ignore** incorrect references to the terms precision and accuracy **Indicative scientific points may include:** **Evaluation of Fig. 22.1** • Comment on the line • The straight line misses one error bar / anomalous point ringed or indicated • Too few data points plotted • The triangle used to calculate the gradient is (too) small • Some plots should have been repeated / checked • No error bars for current • ‘Not regular intervals’ (for current) • No origin shown (AW) **Evaluation of analysis** • The value of B is close to the accepted value • The difference of only 7% • No absolute or percentage uncertainty in B shown (AW) • Worst-fit line or maximum / minimum gradient line could have been used to determine the (absolute or percentage) uncertainty in B • F against I graph should be a straight line or • BL = gradient (any subject) **Examiner’s Comment** This was the second level of response (LoR) question in the paper. It required evaluation of a graph drawn by a student and the analysis shown in the box on page 24. Most candidates realised that the graph had few data points, the triangle used for the gradient was too small and the line drawn totally missed one of the error bars. The analysis shown by the candidate did not include an absolute uncertainty in B, which made the statement written by the student lack credibility. Many candidates wrote about drawing doing a line of worst-fit and determining the percentage uncertainty. This was only possible if there were more

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Question			Answer/Indicative content	Marks	Guidance
					data points and the error bars for the *F* values reduced by perhaps repeating the measurements. Once again, there was a good spread of marks amongst the three levels.
	b	i	There is a changing / fluctuating (magnetic) field / flux (linkage)	M1	**Note:** This changing flux can be anywhere **Allow '**the direction of the field oscillates'
			(magnetic) field / flux (linkage) in <u>core</u> and <u>secondary</u> (coil)	A1	**Allow** 'the core helps to link the flux to the secondary coil'
			Statement of Faraday's law: e.m.f. (induced) ∝ <u>*rate*</u> of change of (magnetic) flux <u>linkage</u>	B1	**Allow** 'equal to / =' **Ignore** 'cutting of flux' **Not** just $E = (–)\Delta(N\varphi)/\Delta t$ **Examiner's Comment** The topic electromagnetic induction always challenges candidates. Successful responses often showed correct use of technical terms such as *magnetic flux* or *flux linkage.* Most candidates scored a mark for correctly stating Faraday's law of electromagnetic induction. Many realised that an alternating current produced an alternating magnetic flux within the iron core and this change in flux produced an e.m.f. at the secondary coil. One of the popular misconceptions was that there was an alternating current (or induced e.m.f.) within the iron-core. A small number of candidates referred to **electro**magnetic field in their descriptions rather than magnetic field.

Question			Answer/Indicative content	Marks	Guidance
		ii	1 (I_S =) 24/12 or 2.0 (A)	C1	
			(I_P =) $\frac{20}{400} \times 2.0$		
			(current in primary =) 0.10 (A)	A1	**Allow** 1 sf answer
			or		
			(V_P =) 12 × 20 or 240 (V)	C1	
			(I_P =) $\frac{24}{240}$		
			(current in primary =) 0.10 (A)	A1	**Allow** 1 sf answer
			2 Idea of changing / increasing (magnetic) field / flux / current (in primary) at the start	B1	**Note:** Any labels used must be clearly defined
			Eventually current and flux (linkage) are constant, therefore no e.m.f.	B1	**Examiner's Comment** This question on current in the primary coil was successfully answered by most candidates. The most favourable method was to calculate the current in the secondary and then the current in the primary coil. The turn-ratio equation and $P = VI$ were effortlessly used to arrive at the correct answer of 0.10 A. Full marks were rarely scored but many top-end candidates did manage to score a mark for suggesting that the lamp was lit for a short period of time at the start because *'there was a changing magnetic flux as the current increased from zero to a steady value'.* Too many answers focussed on the requirement of an alternating supply for an induced e.m.f. in the secondary coil and how a battery is not an alternating supply.
			Total	13	

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Question			Answer/Indicative content	Marks	Guidance
8			Flemings left hand rule / the force on the electron is in the plane of the paper, right angles to the velocity and 'downwards'. Circular path within field in a clockwise direction.	B1 B1	**Note**: If drawn on Fig. 22.1, then judge 'circular' path by eye.
			Total	**2**	
9			A	1	
			Total	**1**	
10			A	1	
			Total	**1**	